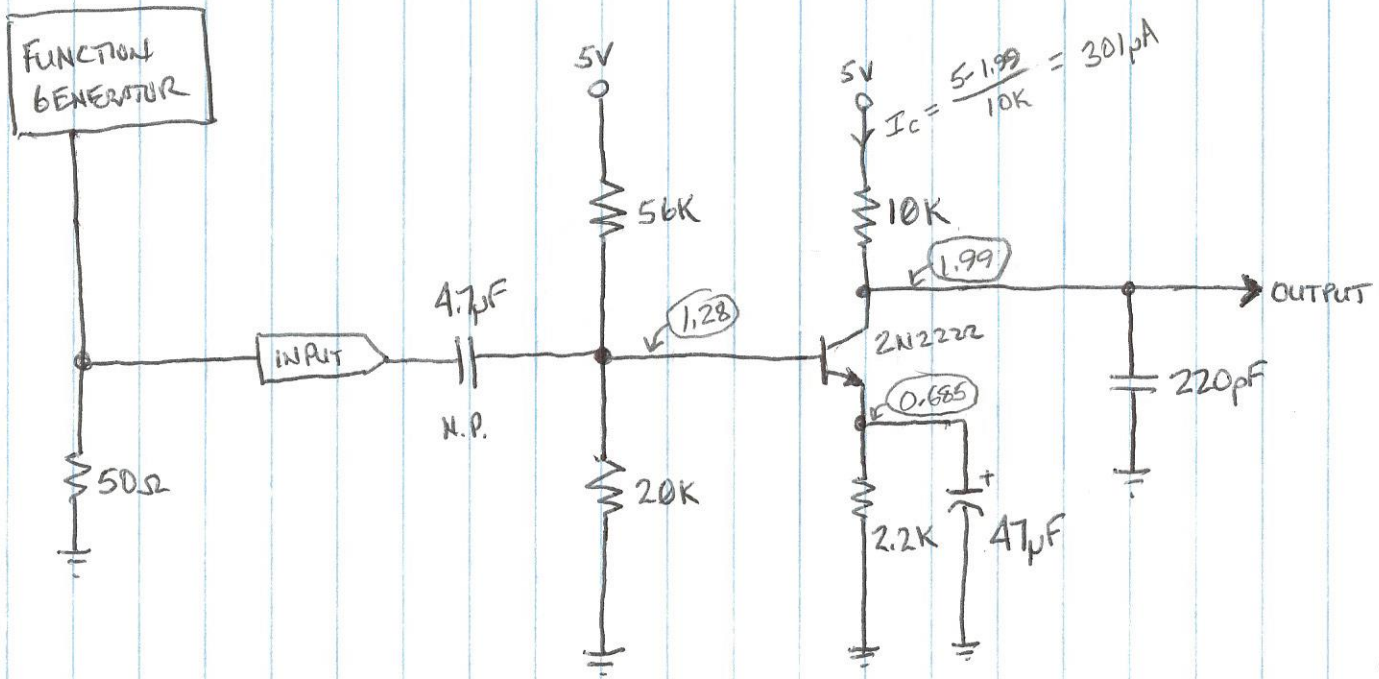
BASIC FREQUENCY RESPONSE OF A COMMON EMITTER AMPLIFIER



$$V_T = \frac{kT}{q} = 26mV \text{ @ ROOM TEMP}$$

$$r_e = 1/g_m$$

$$g_m = \frac{I_c}{V_T} = \frac{301\mu A}{26mV} = 11.6mS$$

$$r_e = 86\Omega$$

$$A_v = -(g_m)(R_c) = 10K * 11.6mS$$

$$= -116$$

$$A_v = -\frac{R_c}{r_e}$$

$$f_{3dB} = \frac{1}{2\pi RC}$$

$$f_H = \frac{1}{2\pi(10K)(220p)} = 72.4KHz$$

$$f_L = \frac{1}{2\pi(47\mu F)(86)} = 39Hz$$